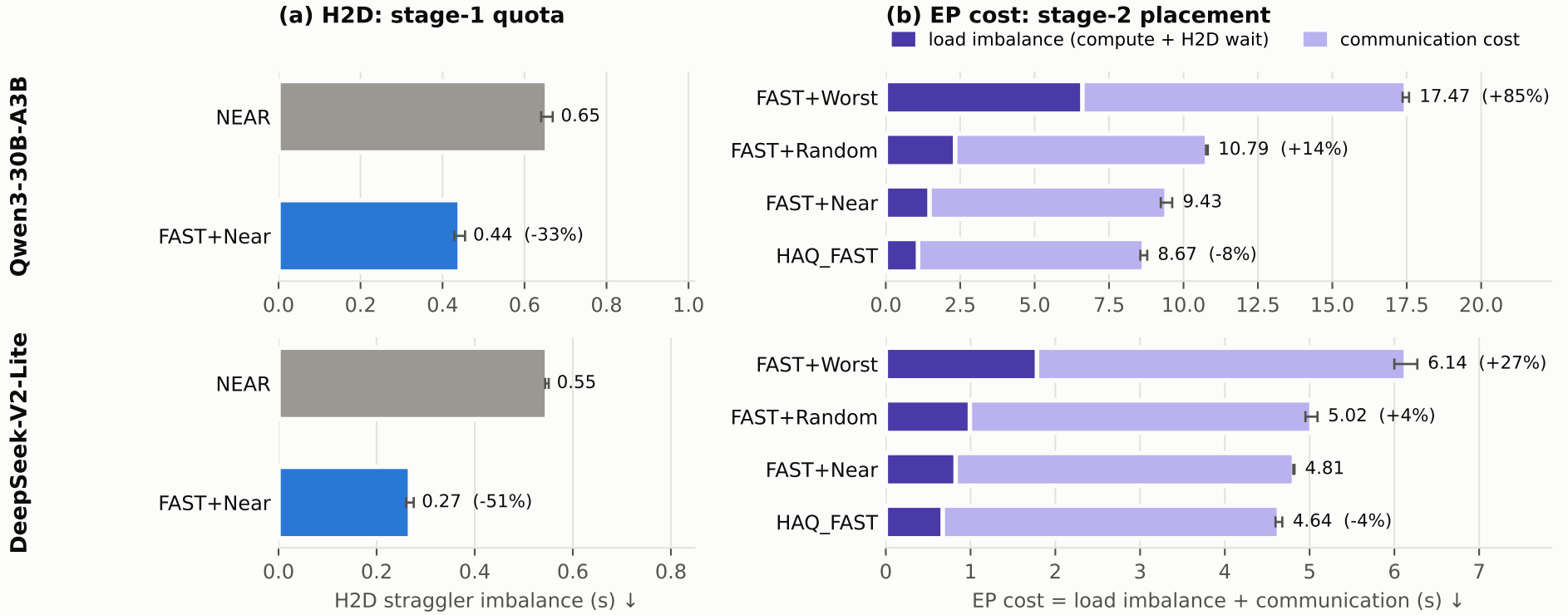
Stage-wise cost breakdown: R8, serial expert execution, inline H2D (decode sums, mean ± std, 3 runs)



(a) relative to NEAR; (b) relative to FAST+Near. HAQ_FAST uses a hit-aware quota; its H2D imbalance stays near FAST (Qwen 0.46 vs 0.44 s, DeepSeek 0.32 vs 0.27 s).